

## Do like Rubi : don't be afraid of using arctanh !

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**In memory of Albert Rich**

### 1 Introduction

In our *Education session* abstract ([1]), we mentioned the desire to modernize the tables of integrals used in mathematics courses. Saying this should come with recalling basic algebra and introducing students to hyperbolic functions. This is why we have decided to propose this *extended* abstract. We want to be as clear as possible and provide all relevant details. We are well aware that introducing hyperbolic functions will require more time than usual. But it will be well worth it. So here are the prerequisites for students if they wish to use the table of integrals proposed in these pages. When the definitions of the hyperbolic sine, cosine and tangent functions have been given, the following results should be presented to the students:

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = 1 - \frac{2}{e^{2x} + 1}$$

where we have used a "proper fraction" to get the last equality. The trigonometric identities

$$\cos^2(x) + \sin^2(x) = 1 \quad \text{and} \quad 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$$

have their corresponding hyperbolic ones where the "+" sign changes for a "-" sign:

$$\cosh^2(x) - \sinh^2(x) = 1 \quad \text{and} \quad 1 - \tanh^2(x) = \frac{1}{\cosh^2(x)}.$$

The function  $x \mapsto \tanh(x)$  is a one-to-one correspondance from  $\mathbb{R}$  to the open interval  $] -1, 1[$  whose derivative is (using the quotient rule and the identities)

$$\frac{d}{dx}(\tanh(x)) = \frac{1}{\cosh^2(x)}.$$

To find the inverse hyperbolic tangent – the arctanh function –, we interchange the variables  $x$  and  $y$  and set  $x = \tanh(y)$ . Solving for  $y$ , we have

$$x = 1 - \frac{2}{e^{2y} + 1} \iff e^{2y} = \frac{1+x}{1-x}.$$

We take the natural log on both sides, not forgetting that  $x$  should stay between  $-1$  and  $1$  :

$$y = \operatorname{arctanh}(x) \equiv \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad (-1 < x < 1).$$

Students can compute the derivative of  $\operatorname{arctanh}(x)$  exactly in the same way they did it for the derivative of  $\arctan(x)$ . We have  $y = \operatorname{arctanh}(x)$  and we want  $\frac{dy}{dx}$ . Using the equality  $x = \tanh(y)$  implies

$$\frac{dx}{dy} = \frac{1}{\cosh^2(y)} = \frac{\cosh^2(x) - \sinh^2(x)}{\cosh^2(y)} = 1 - \tanh^2(y) \implies \frac{dy}{dx} = \frac{1}{1 - \tanh^2(y)} = \frac{1}{1 - x^2}.$$

A nice symmetric formula is ready to appear in an upcoming table of integrals because

$$\frac{d}{dx}(\operatorname{arctan}(x)) = \frac{1}{1+x^2} \quad \text{and} \quad \frac{d}{dx}(\operatorname{arctanh}(x)) = \frac{1}{1-x^2}.$$

Having used hyperbolic functions in their calculus course, students will find it easier in their next differential equations course. Today, modern ODE textbooks contain intensive use of computer algebra systems and these systems are complex oriented. For example, consider a problem like a falling object where the air resistance force is proportional to the square of the velocity; the solution involves the hyperbolic tangent function as figure 1 shows. In TI-Nspire CAS,  $i$  stands for the imaginary unit.

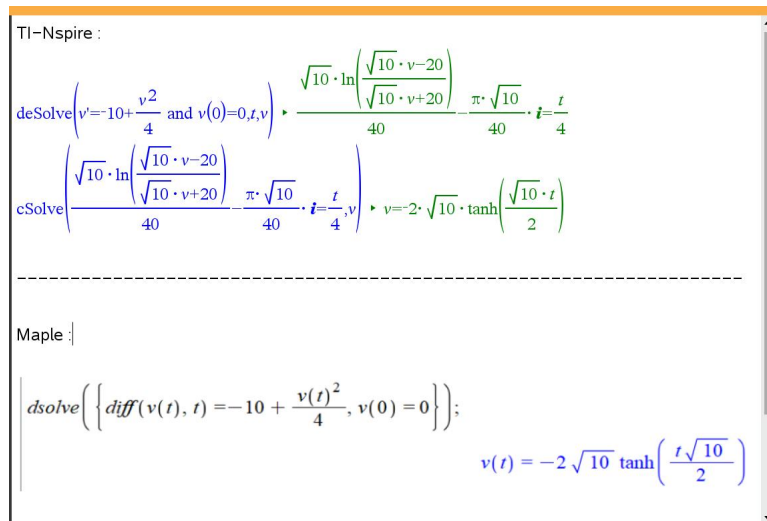


Figure 1: Solution of an ODE with TI-Nspire CAS and Maple.

## 2 The first integration formulas

Regarding a new table of integrals, we wrote it was time to stop inserting in tables  $\ln(|x|) + C$  as an antiderivative of  $\frac{1}{x}$ ; time to use symmetry between formulas wherever it is possible; time to use an antiderivative even though it is non-real over a certain interval but looks simpler than the usual ones found in tables; and time to look for a continuous primitive on (at least) the same domain as the integrand. We are aware that choices must be made if we need to select a single antiderivative. One of these choices, we indicated, is to decide to use the hyperbolic arctangent function. Rubi (the Rule-based Integrator system) does this. With Wolfram Alpha (accessible on a smartphone), and with many computer algebra systems on the market, we are conscious that there is no need for a *new* table of integrals! But we think we still need to respond to the textbooks who have been ignoring the contribution from the CAS over the last 35 years. On a personal side, this is my way to thank **Derive** and Rubi. Here are the first 8 integration formulas that should be inserted into a table of integrals. Note we don't need to include the integral of the exponential expression  $b^x$  ( $b > 0, b \neq 1$ ) because we want less formulas in the tables and more memorization by the students. Anyway,  $b^x = e^{x \ln(b)}$ .

|                                                                                                        |                                                                                                         |
|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| (1) $\int dx = x$                                                                                      | (2) $\int x^n dx = \frac{x^{n+1} - 1}{n + 1} \quad (n \neq -1)$                                         |
| (3) $\int \frac{1}{x} dx = \ln(x)$                                                                     | (4) $\int e^x dx = e^x$                                                                                 |
| (5) $\int \sin(x) dx = -\cos(x)$                                                                       | (6) $\int \cos(x) dx = \sin(x)$                                                                         |
| (7) $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \text{arctan}\left(\frac{x}{a}\right) \quad (a \neq 0)$ | (8) $\int \frac{1}{a^2 - x^2} dx = \frac{1}{a} \text{arctanh}\left(\frac{x}{a}\right) \quad (a \neq 0)$ |

Table 1: Eight basic integral formulas students should remember

In this table, five formulas out of eight are the ones found in every textbook. Let us explain our choices for the three others. Formula (2) was chosen as a wink to **Derive**: the constant of integration  $-\frac{1}{n+1}$  has always been added by **Derive** in his integral of  $x^n$ . The reason: if we let  $n$  approaches  $-1$ , the right hand side of formula (2) simplifies into  $\ln(x)$  yielding formula (3). This is the only formula where a "Kahanian constant of integration" has been added and, of course, students won't need it when asked to compute a particular definite integral involving a power expression.

Formula (3) sends the following message: if you are looking at the tables of integrals contained in many calculus textbooks, the antiderivative chosen for the expression  $\frac{1}{x}$  is  $\ln(|x|) + C$ . Both  $\ln(|x|)$  and  $\ln(-x)$  are valid antiderivatives of  $\frac{1}{x}$  as derivation shows. The problem is when textbook authors write  $\int \frac{1}{x} dx = \ln(|x|) + C$ . They seem to forget the fact that the expression  $\frac{1}{x}$  is not continuous on any interval containing the origin. When comes time to choose an antiderivative for  $\frac{1}{x}$ , every CAS is using  $\ln(x)$  – for TI-Nspire, for

example you need to set "Real or Complex Format" to "Rectangular". Why computer algebra implementers abhor  $\ln(|x|)$ ? Very good explanations have been given many years ago, see [2]. Finally, formula (8) was chosen to recall the huge importance Rubi has had – and will continue to have – on symbolic integration, namely by its desire to use symmetry when possible, even though  $\operatorname{arctanh}(x)$  is real only for  $-1 < x < 1$ .

## 2.1 Using table 1 to compute integrals

Students – especially engineering ones who have been using tables for a long time – will probably replace formulas (3) to (6) by ones containing a parameter  $a$ . So, for  $a \neq 0$ , they can easily remember these formulas :

$$\int \frac{1}{x+a} dx = \ln(x+a) \quad \text{and} \quad \int e^{ax} dx = \frac{e^{ax}}{a};$$

$$\int \sin(ax) dx = -\frac{\cos(ax)}{a} \quad \text{and} \quad \int \cos(ax) dx = \frac{\sin(ax)}{a}.$$

Let's give some examples of using Table 1 assuming that students have also learned the basic integration techniques of substitution and integration by parts. Our table is already very different from the usual tables. A system like Rubi is based on rules and can served as a *teaching assistant*. Why not take the opportunity to ask questions differently? The following examples point in this direction.

### 2.1.1 Example 1

Consider  $\int_{-5}^{-4} \frac{1}{x-3} dx$ . This is a definite integral where the integrand is continuous over the interval of integration. The solution can always consist on using  $\ln(x-3)$  for the antiderivative if your students are comfortable with this, or if the system they use allows intermediate complex values in the computation (because complex numbers will cancel). If your students are confused by logs of negative numbers, use  $\ln(|x-3|)$  or  $\ln(3-x)$  as antiderivatives. Everyone will get the same answer :  $\ln(\frac{7}{8})$ .

### 2.1.2 Example 2

Consider  $\int \frac{1}{x^2+6x+8} dx$ . If we set  $u = x+3$ , we have  $du = dx$  and

$$\int \frac{1}{x^2+6x+8} dx = \int \frac{1}{(x+3)^2-1} dx = \int \frac{1}{u^2-1} du = -\operatorname{arctanh}(x+3).$$

Because it was not a perfect square, Rubi (4.16.1) yields the answer  $\frac{1}{2} \ln(x+2) - \frac{1}{2} \ln(x+4)$  which has, of course, a different real domain than the one with the  $\operatorname{arctanh}$ .

More generally, remind your students of the importance of the sign of  $b^2 - 4ac$  in the quadratic expression  $ax^2 + bx + c$ . Ask them to practice the completion of square and

choose an appropriate substitution. Then they should be able to decide which formula among (2), (7) and (8) they should use in order to compute an integral like

$$\int \frac{1}{s^2 + \alpha s + \beta} ds \quad (\alpha, \beta \neq 0).$$

### 2.1.3 Example 3

Consider  $\int \frac{x}{x^2+6x+8} dx$ . Because we don't have a multiple of  $x+3$  on the numerator, a change of variable is not appropriate. We also don't complete the square of the denominator –the roots of the denominator are real and this could lead to an unnecessarily complicated antiderivative. A simple partial fraction expansion shows that  $\frac{x}{x^2+6x+8} = \frac{2}{x+4} - \frac{1}{x+2}$  and this leads to

$$\int \frac{x}{x^2+6x+8} dx = 2\ln(x+4) - \ln(x+2).$$

More generally, the students should be able to compute easily any integral of the form

$$\int \frac{Ax+B}{Cx^2+Dx+E} dx.$$

### 2.1.4 Example 4

Consider  $I = \int \frac{1}{x^4-1} dx$ . Let your students having access to a CAS command such as `expand()`. They will see that

$$\frac{1}{x^4-1} = \frac{1}{4(x-1)} - \frac{1}{4(x+1)} - \frac{1}{2(x^2+1)} \implies I = \frac{\ln(x-1)}{4} - \frac{\ln(x+1)}{4} - \frac{\arctan(x)}{2}.$$

Instead of expanding, Rubi uses a rule detecting the equality  $\frac{1}{x^4-1} = -\frac{1}{2(1+x^2)} - \frac{1}{2(1-x^2)}$  and yields a more compact answer involving an `arctan` and an `arctanh` :

$$\int \frac{1}{x^4-1} dx = -\frac{\arctan(x)}{2} - \frac{\operatorname{arctanh}(x)}{2}.$$

Here is a question to the students: use what the partial fractions expansion command yielded and show this can be converted into Rubi's answer.

### 2.1.5 Example 5

A long time ago, calculus textbooks contained questions like the following : use integration by parts twice to compute the integral of expressions like  $x^2 \sin(3x)$  and  $e^{ax} \sin(bx)$ . Also, a large section of the course textbook was on integrals of powers of trigonometric functions. With CAS availability, these kinds of questions are now rarer. Our suggestion is: stick only to simple cases and ask students to then demonstrate reduction formulas like the followings.

$$\int \sin^n(x) dx = -\frac{\sin^{n-1}(x) \cos(x)}{n} + \frac{n-1}{n} \int \sin^{n-2}(x) dx \quad (n \in \mathbb{N}, n \geq 2);$$

$$\int x^m \sin(ax+b) dx = -\frac{x^m \cos(ax+b)}{a} + \frac{m}{a} \int x^{m-1} \cos(ax+b) dx \quad (m \in \mathbb{N}, a \neq 0).$$

We can also create new types of questions that will make use of the arctanh function such as showing that

$$\int \frac{\sin(2x)}{1 - \sin^4(x)} dx = \operatorname{arctanh}(\sin^2(x)).$$

### 2.1.6 Example 6

According to David Jeffrey's abstract for the ACA2023 in Warsaw (see [3]), the upcoming version of Rubi will show that

$$\int \frac{\sin(ax) + \sin(bx)}{\cos(ax) + \cos(bx)} dx = -\frac{2}{a+b} \ln\left(\cos\left(\frac{(a+b)x}{2}\right)\right).$$

What are the rules Albert have used for deducing this formula? We can certainly try to show the formula by using some trigonometric identities. Here, we show another illustration of the benefit Rubi has on Education. Start with the usual trigonometric identities

$$\sin(x \pm y) = \sin(x) \cos(y) \pm \cos(x) \sin(y) \quad \text{and} \quad \cos(x \pm y) = \cos(x) \cos(y) \mp \sin(x) \sin(y)$$

and deduce the following:

$$\frac{\sin(x+y) + \sin(x-y)}{2} = \sin(x) \cos(y) \quad \text{and} \quad \frac{\cos(x+y) + \cos(x-y)}{2} = \cos(x) \cos(y).$$

Dividing side by side the last two equalities yields

$$\frac{\sin(x+y) + \sin(x-y)}{\cos(x+y) + \cos(x-y)} = \tan(x)$$

which is equivalent to

$$\frac{\sin(\alpha) + \sin(\beta)}{\cos(\alpha) + \cos(\beta)} = \tan\left(\frac{\alpha + \beta}{2}\right).$$

This shows that

$$\frac{\sin(ax) + \sin(bx)}{\cos(ax) + \cos(bx)} = \tan\left(\frac{(a+b)x}{2}\right).$$

Finally, the usual formula (9) shown in the upcoming table 2 finishes the job.

## 3 The remaining basic trigonometric integrals

The tables of integrals used in textbooks usually contain the 8 additional formulas shown in table 2. Formulas (13) to (16) are the usual ones, these are the same everywhere. Because there is no more absolute value inserted into formula (3) of table 1, the same occurs in formulas (9) and (10). Formulas (11) and (12) are using arctanh in their antiderivative like Rubi does.

|                                                          |                                                           |
|----------------------------------------------------------|-----------------------------------------------------------|
| (9) $\int \tan(x) dx = -\ln(\cos(x))$                    | (10) $\int \cot(x) dx = \ln(\sin(x))$                     |
| (11) $\int \sec(x) dx = \operatorname{arctanh}(\sin(x))$ | (12) $\int \csc(x) dx = -\operatorname{arctanh}(\cos(x))$ |
| (13) $\int \sec^2(x) dx = \tan(x)$                       | (14) $\int \csc^2(x) dx = -\cot(x)$                       |
| (15) $\int \sec(x) \tan(x) dx = \sec(x)$                 | (16) $\int \csc(x) \cot(x) dx = -\csc(x)$                 |

Table 2: The remaining basic trigonometric integrals

The expression  $\sec(x)$  is discontinuous at every point  $x$  of the form  $x = \frac{(2n-1)\pi}{2}$  where  $n \in \mathbb{Z}$ . So, for  $-\frac{\pi}{2} < x < \frac{\pi}{2}$ , the simple one-to-one change of variables  $u = \sin(x)$ , the first trigonometric identity and formula (8) easily prove (11) because

$$\int \sec(x) dx = \int \frac{\cos(x)}{1 - \sin^2(x)} dx = \int \frac{1}{1 - u^2} du = \operatorname{arctanh}(\sin(x)).$$

A similar proof works for formula (12). Note that the textbooks' usual antiderivative in formula (12) is  $\ln(|\sec(x) + \tan(x)|)$  which is exactly the same as  $\operatorname{arctanh}(\sin(x))$ . Popular CAS often use  $\ln(\sec(x) + \tan(x))$  for their antiderivative of  $\sec(x)$  which is the same expression as  $\ln(\tan(\frac{2x+\pi}{4}))$ .

## 4 The remaining formulas : sums and differences of squares under a square root

We conclude our *new* table of integrals by adding usual formulas involving square roots of quadratic expressions of the form  $x^2 + a^2$ ,  $x^2 - a^2$  and  $a^2 - x^2$ . Our choices are again motivated by Rubi: in fact, formulas (17), (18) and (19) of table 3 are exactly those returned. We will show how to deduce these. Of course, differentiation would work but a good exercise for students would be to use their trigonometric identities in order to compute each integral.

When this will be accomplished, then formulas (20), (21) and (22) will simply be a consequence of formula 87 of page 417 ([4]):

$$\int (a + b x^n)^p dx = \frac{x (a + b x^n)^p}{n p + 1} + \frac{n p a}{n p + 1} \int (a + b x^n)^{p-1} dx.$$

More precisely, formulas (20), (21) and (22) will be obtained from formulas (17), (18) and (19) by using the values  $n = 2$  and  $p = \frac{1}{2}$  and choosing appropriate values for  $a$  and  $b$  (sometimes  $a^2$ ,  $-a^2$ ,  $1$ ,  $-1$ ).

|      |                                                                                                              |
|------|--------------------------------------------------------------------------------------------------------------|
| (17) | $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arctan\left(\frac{x}{\sqrt{a^2 - x^2}}\right)$                        |
| (18) | $\int \frac{1}{\sqrt{a^2 + x^2}} dx = \operatorname{arctanh}\left(\frac{x}{\sqrt{a^2 + x^2}}\right)$         |
| (19) | $\int \frac{1}{\sqrt{x^2 - a^2}} dx = \operatorname{arctanh}\left(\frac{x}{\sqrt{x^2 - a^2}}\right)$         |
| (20) | $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \int \frac{1}{\sqrt{a^2 - x^2}} dx$ |
| (21) | $\int \sqrt{a^2 + x^2} dx = \frac{x}{2} \sqrt{a^2 + x^2} + \frac{a^2}{2} \int \frac{1}{\sqrt{a^2 + x^2}} dx$ |
| (22) | $\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \int \frac{1}{\sqrt{x^2 - a^2}} dx$ |

Table 3: Sums and differences of squares under a square root ( $a \neq 0$ )

Let us recall that if  $a \in \mathbb{R}$  is such that  $a \neq 0$ , then the signum function is defined by

$$\operatorname{sign}(a) = \begin{cases} 1, & a > 0 \\ -1, & a < 0 \end{cases}.$$

The following are trivial :  $\operatorname{sign}(a) = \frac{|a|}{a} = \frac{1}{\operatorname{sign}(a)}$  and if  $f$  is an odd function ( $f(-x) = -f(x)$ ), then  $\operatorname{sign}(a) f(x) = f(\operatorname{sign}(a) x)$ . In order to demonstrate formulas (17) to (19), we will use some trigonometric substitutions. Figure 2 proves the following equality for  $0 < y < 1$  :

$$\arcsin(y) = \arctan\left(\frac{y}{\sqrt{1-y^2}}\right).$$

And because  $\arcsin$  and  $\arctan$  are odd functions, the equality is also true for  $-1 < y < 0$ .

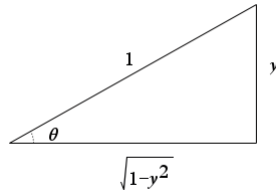


Figure 2: A right triangle where  $\sin(\theta) = y$  and  $\tan(\theta) = \frac{y}{\sqrt{1-y^2}}$

We can now demonstrate formula (17) with  $a \neq 0$ . We use the one to one substitution  $x = a \sin(\theta)$ ,  $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$  so that  $dx = a \cos(\theta) d\theta$ . The cosine function being positive in the interval, we have  $\sqrt{a^2 - x^2} = \sqrt{a^2 \cos^2(\theta)} = |a| \cos(\theta)$  and we use the fact that the arcsine function is an odd function :

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \int \frac{a \cos(\theta)}{|a| \cos(\theta)} d\theta = \text{sign}(a) \theta = \text{sign}(a) \arcsin\left(\frac{x}{a}\right) = \arcsin\left(\frac{x}{|a|}\right).$$

We get rid of the absolute value by using an arctan instead of an arcsin as the following computation shows:

$$\arcsin\left(\frac{x}{|a|}\right) = \arctan\left(\frac{x/|a|}{\sqrt{1 - (x/|a|)^2}}\right) = \arctan\left(\frac{x/|a|}{\sqrt{a^2 - x^2}/|a|}\right) = \arctan\left(\frac{x}{\sqrt{a^2 - x^2}}\right).$$

So formula (17) has been proven and we showed that when  $a$  is positive,  $\arcsin(\frac{x}{a})$  can also be used as antiderivative of  $\frac{1}{\sqrt{a^2 - x^2}}$  (and this will be the case when  $a$  is not symbolic but numeric).

Let us now prove formula (18). Here we use  $x = a \tan(\theta)$  with  $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ . The secant function is also positive in the interval, so  $\sqrt{a^2 + x^2} = |a| \sec(\theta)$ . We compute the derivative and get  $dx = a \sec^2(\theta) d\theta$ . Formula (11) of table 5 yields

$$\int \frac{1}{\sqrt{a^2 + x^2}} dx = \int \frac{a \sec^2(\theta)}{\sqrt{a^2 + a^2 \tan^2(\theta)}} d\theta = \int \frac{a \sec(\theta)}{|a|} d\theta = \text{sign}(a) \text{arctanh}(\sin(\theta)).$$

Using the fact that the inverse hyperbolic tangent function is odd, we have

$$\text{sign}(a) \text{arctanh}(\sin(\theta)) = \text{arctanh}\left(\frac{a \tan(\theta)}{|a| \sec(\theta)}\right) = \text{arctanh}\left(\frac{x}{\sqrt{a^2 + x^2}}\right).$$

Alternative antiderivatives – possibly differing by a constant – for the integrand  $\frac{1}{\sqrt{a^2 + x^2}}$  are  $\text{arcsinh}\left(\frac{x}{a}\right)$  when  $a > 0$  and  $\ln(x + \sqrt{a^2 + x^2})$  for every  $a \neq 0$ .

**Remark:** Figure 3 shows that Rubi is not using a trigonometric substitution in order to deduce the formulas of table 3. Instead a different change of variables that get rid of the square root. This makes Rubi an *integrator assistant*. This also shows that a *new* way of teaching integration techniques to calculus students could be looking at the rules Rubi is using.

To prove formula (19), we set  $x = a \sec(\theta)$  which is one to one in the interval  $0 < \theta < \frac{\pi}{2}$ . The function  $\tan(\theta)$  is positive in this interval, so  $\sqrt{x^2 - a^2} = |a| \tan(\theta)$ . We have

$$dx = a \sec(\theta) \tan(\theta) d\theta \quad \text{and} \quad \sqrt{x^2 - a^2} = \sqrt{a^2 \sec^2(\theta) - a^2} = \sqrt{a^2 \tan^2(\theta)} = |a| \tan(\theta).$$

We repeat what we did for formula (18) and use again formula (11) of table 2:

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \int \frac{a \sec(\theta) \tan(\theta) d\theta}{|a| \tan(\theta)} = \int \text{sign}(a) \sec(\theta) d\theta = \text{sign}(a) \text{arctanh}(\sin(\theta)).$$

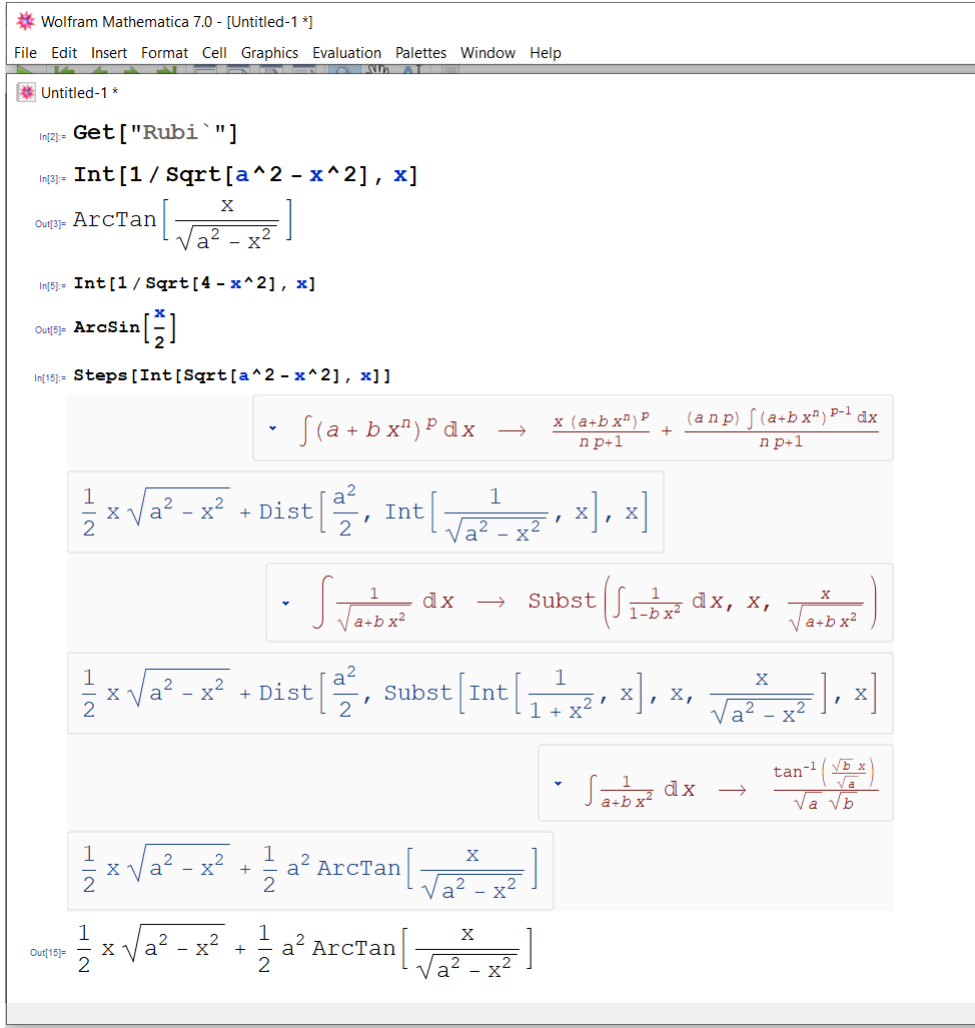


Figure 3: Rubi (version 4.16.1) showing the steps

Using again the fact the inverse hyperbolic tangent function is odd, we have

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \text{arctanh}\left(\frac{|a|}{a} \sin(\theta)\right) = \text{arctanh}\left(\frac{|a| \tan(\theta)}{a \sec(\theta)}\right) = \text{arctanh}\left(\frac{\sqrt{x^2 - a^2}}{x}\right).$$

The last expression is a valid antiderivative whose domain is the same as the integrand, namely  $|x| > a$ . But a direct computation shows that

$$\frac{d}{dx} \left( \text{arctanh}\left(\frac{x}{\sqrt{x^2 - a^2}}\right) \right) = \frac{1}{\sqrt{x^2 - a^2}}.$$

So even if the real domain of  $\text{arctanh}\left(\frac{x}{\sqrt{x^2 - a^2}}\right)$  is the empty set, our desire of symmetry

implies to choose it as our antiderivative of  $\frac{1}{\sqrt{x^2 - a^2}}$ . Finally, an alternative antiderivative for the integrand  $\frac{1}{\sqrt{x^2 - a^2}}$  is  $\ln(x + \sqrt{x^2 - a^2})$  for every  $a \neq 0$ .

## 5 Conclusion

On the Rubi website, the late Albert Rich wrote: "Indefinite integration accounts for only a tiny fraction of all mathematical knowledge". His legacy will certainly include a reflection on what every calculus professor should do. Only for this, I think we all gained something. On a personal side, this reflection about the tables of integrals was an opportunity for me to continue what I have been talking about in the last 3 ACA conferences : to think about the curriculum in mathematics courses – not only in calculus courses – and try to make it more interesting to students who have access to tools everywhere. In the paper [6], David Stoutemyer wrote: "This article address the usefulness of computer algebra systems as productivity tools to help amateur users accomplish their tasks without necessarily being aware of the underlying algorithms, transformation functions and their nomenclature". What we teach and how we do it must be clarified more than ever if we want to... keep our jobs!

### Keywords

Calculus textbooks, Tables of integrals, Computer Algebra Systems

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